

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Circles	Hard

Question

The equation $x^2 + (y - 1)^2 = 49$ represents circle A. Circle B is obtained by shifting circle A down **2** units in the xy -plane. Which of the following equations represents circle B?

Answer

- A. $(x - 2)^2 + (y - 1)^2 = 49$
- B. $x^2 + (y - 3)^2 = 49$
- C. $(x + 2)^2 + (y - 1)^2 = 49$
- D. $x^2 + (y + 1)^2 = 49$

Correct Answer: D

Rationale

Choice D is correct. The graph in the xy -plane of an equation of the form $(x - h)^2 + (y - k)^2 = r^2$ is a circle with center (h, k) and a radius of length r . It's given that circle A is represented by $x^2 + (y - 1)^2 = 49$, which can be rewritten as $x^2 + (y - 1)^2 = 7^2$. Therefore, circle A has center $(0, 1)$ and a radius of length **7**. Shifting circle A down two units is a rigid vertical translation of circle A that does not change its size or shape. Since circle B is obtained by shifting circle A down two units, it follows that circle B has the same radius as circle A, and for each point (x, y) on circle A, the point $(x, y - 2)$ lies on circle B. Moreover, if (h, k) is the center of circle A, then $(h, k - 2)$ is the center of circle B. Therefore, circle B has a radius of **7** and the center of circle B is $(0, 1 - 2)$, or $(0, -1)$. Thus, circle B can be represented by the equation $x^2 + (y + 1)^2 = 7^2$, or $x^2 + (y + 1)^2 = 49$.

Choice A is incorrect. This is the equation of a circle obtained by shifting circle A right **2** units.

Choice B is incorrect. This is the equation of a circle obtained by shifting circle A up **2** units.

Choice C is incorrect. This is the equation of a circle obtained by shifting circle A left **2** units.